Problem Pack: MATHEMATICS 8 • UNIT 2 REVIEW, A. Exponent Laws and Perfect Squares

Question 1a

Problem: 1. Simplify each expression. Your final answer should contain only positive exponents.
a) x^4 x^7

Student Work:
x^{11}

Feedback

Question 1b

Problem: 1. Simplify each expression. Your final answer should contain only positive exponents.
b) a^{12} ÷ a^5

Student Work:
a^7

Feedback

Question 1c

Problem: 1. Simplify each expression. Your final answer should contain only positive exponents.
c) $(m^{3})4$

Student Work:
m^{12}

Feedback

Question 1d

Problem: 1. Simplify each expression. Your final answer should contain only positive exponents.
d) $(2pq^{2})3$

Student Work:
$8p^{3q}6$

Feedback

Question 1e

Problem: 1. Simplify each expression. Your final answer should contain only positive exponents.
e) (3x^2 / 2y)^2

Student Work:
$9x^{4}⁄_{4y}2$

Feedback

Question 1f

Problem: 1. Simplify each expression. Your final answer should contain only positive exponents. f) 7r^0

Student Work:

7

Feedback

In question 1f, you made a mistake on the term with the zero exponent. You might think that any number multiplied by zero stays itself, but remember that any non-zero base raised to the power of zero is actually equal to one. Try applying that rule to the number outside the variable to find your new total.

Question 2a

Problem: 2. Simplify completely. Use any exponent laws you need, and leave only positive exponents. a) [(2m^{3n}2)$^{2(4mn}$3)] / (8m^2n)

Student Work:

4m^{6n}4 4mn^3 / 8m^2n = 16m^{7n}7 / 8m^2n = 2m^{5n}6

Feedback

Question 2b

Problem: 2. Simplify completely. Use any exponent laws you need, and leave only positive exponents. b) (3x$^{2y)}$3 / (9x^{4y}2)

Student Work:

27x^{6y}3 / 9x^{4y}2 = 3x^2y

Feedback

Question 3

Problem: 3. Classify each number as a perfect square or non-perfect square. If it is a perfect square, give its principal square root. 225 30 4/9 49/20 0.16 0.6

Student Work:

Perfect square: 15 Non-perfect square 2/3, perfect square not perfect square 0.4, perfect square non-perfect square

Feedback

Question 4a

Problem: 4. State how many real square roots each number has. Briefly explain. a) 81

Student Work:
2,

Feedback

Question 4b
Problem: 4. State how many real square roots each number has. Briefly explain. b) 0

Student Work:
1

Feedback

Question 4c
Problem: 4. State how many real square roots each number has. Briefly explain. c) −49

Student Work:
1, −7

Feedback
In question 4c, you went wrong on the number of real square roots for a negative number. You might think negative numbers can have a square root just like positive ones, but multiplying two negative numbers always gives a positive result. Try thinking about what number multiplied by itself could ever equal a negative value.

Question 5
Problem: 5. Evaluate √49. Then list all solutions of x^2 = 49. Explain why the answers are different.

Student Work:
√49 = 7 √x^2 = √49 x = Because we don't know if x is −7 or +7. Whereas, we know it's +7, because it +49

Feedback